LECTURE: 2 BASIC OF C++ (PART - 2)

Mondira Chakraborty

Lecturer, CSE Dept.

Notre Dame University Bangladesh

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What is <bits/stdc++.h>?

- It is basically a header file that includes every standard library.
- In programming contests, people do focus more on finding the algorithm to solve a problem than on software engineering.
- So, using this file is a good idea, when you want to reduce the time wasted in doing chores; especially when your rank is time sensitive.
- This also reduces all the chores of writing all the necessary header files.
- You don't have to remember all the STL of GNU C++ for every function you use.

Mathematical Functions

	Function	Description	Parameter	Result
1	sin(x)	It computes the sine of x.	Double(radian)	Double
2	cos(x)	It computes the cosine of x.	Double(radian)	Double
3	tan(x)	It computes the tangent of x.	Double(radian)	Double
4	acos(x)	It finds the inverse cosine of x.	Double(radian)	Double
5	asin(x)	It finds the inverse sine of x.	Double(radian)	Double
6	atan(x)	It finds the inverse tangent of x.	Double(radian)	Double
7	abs(x)	Returns absolute value of the argument.	Int	Int
8	ceil(x)	Returns the smallest whole number that is not less than x.	Double	Double
9	floor(x)	Returns the largest whole number that is not greater than x.	Double	Double

Mathematical Functions

	Function	Description	Parameter	Result
10	pow(x, y)	Returns x^y, if x is a negative number, y must be a whole number	Double	Double
11	sqrt(x)	It computes the square root of x.	Int, Double	Double
12	exp(x)	It computes the exponential e raised to the power x.	Double	Double
13	log(x)	It computes the natural logarithm of x.	Double	Double
14	log10(x)	It computes the common logarithm of x.	Double	Double
15	round(x)	It rounds off the value of x.	Double	Double
16	isgreater(x,y)	It determines whether x is greater than y or not.	Same Type	1, 0
17	isgreaterequal(x,y)	It determines whether x is greater than or equal to y or not.	Same Type	1, 0

Mathematical Functions

	Function	Description	Parameter	Result
18	less(x,y)	It determines whether x is less than y or not.	Same Type	1, 0
19	islessequal (x,y)	It determines whether x is less than or equal to y.	Same Type	1, 0
20	rand()	Use to generate a random number	-	library-dependent (at least 32767)
21	srand()	Act as a seed value, randomize the result of rand()	-	-

Mathematical Functions (Example)

```
//generate random numbers example

#include<iostream>
#include<cstdlib> //For rand and srand
#include<ctime> //For the time function

using namespace std;

int main()
  {
      //Get the system time
      unsigned seed = time(0);
      //Seed the random number genrator
      srand(seed);

      int maxValue = 50, minValue = 12;

      //Display three random numbers.

      for(int i = 0; i < 10; i++)
            cout << rand() % ( maxValue - minValue + 1 ) + minValue << endl;

  return 0;
  }
```

What is String?

- In C++, string is an object of **std::string** class that represents sequence of characters
- We can perform many operations on strings such as concatenation, comparison, conversion etc.
- A string variable contains a collection of characters surrounded by double quotes.

String Example

```
#include <iostream>
using namespace std;
int main( ) {
    string s1 = "Hello";
        char ch[] = { 'C', '+', '+'};
        string s2 = string(ch);
        cout<<s1<<endl;
        cout<<s2<<endl;
}
```

Output:

```
Hello
C++
```

String Input/Output

- We can use both cin and getline() function to take inputs from keyboard.
- But, cin considers a space (whitespace, tabs, etc) as a terminating character, so it can only store a single word (even if you type many words)
- That's why when we want to take a text input, we use getline().

String I/O Example

```
string fullName;
cout << "Type your full name: ";
cin >> fullName;
cout << "Your name is: " << fullName;

// Type your full name: John Doe
// Your name is: John
```

```
string fullName;
cout << "Type your full name: ";
getline (cin, fullName);
cout << "Your name is: " << fullName;

// Type your full name: John Doe
// Your name is: John Doe
```

- The left code is expected to print "John Doe", but it only prints "John".
- That's why, when working with strings, we often use the getline() function to read a line of text

What is String Function?

- A string is referred to as an array of characters.
- It is one of the most fundamental datatypes in C++ and it comes with a huge set of inbuilt functions.
- These functions are important for problem solving or programming contest.
- Some basic functions are discussed and we will try to solve some problems regarding this.

String Functions

- <u>String Length – length() or size()</u>

```
string txt = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
cout << "The length of the txt string is: " << txt.length();

string txt = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
cout << "The length of the txt string is: " << txt.size();
```

String Functions

- Accessing characters of a string variable

```
string myString = "Hello";
cout << myString[0];
// Outputs H
```

```
string myString = "Hello";
cout << myString[1];
// Outputs e
```

```
#include <iostream>
#include <string>
using namespace std;

int main() {
  string myString = "Hello";
  cout << myString.at(0);
  return 0;
}
```

String Functions

- Concatenation (Add two or more strings)

```
string firstName = "John";
string lastName = "Doe";
string fullName = firstName + " " + lastName;
cout << fullName;


string firstName = "John ";
string lastName = "Doe";
string fullName = firstName.append(lastName);
cout << fullName;
```

String Functions

- <u>String comparison</u>

```
#include <iostream>
#include <string>
using namespace std;

int main () {
    string str1 = "Code!";
    string str2 = "code!";
        if (str1 == str2) {
            cout << "Strings are equal";
}
else {
        cout << "Strings are not equal";
}
  return 0;
}
```

```
#include <iostream>
#include <string>
using namespace std;

int main () {
    string str1 = "Poet";
    string str2 = "Poem";
    int Comp_result = str1.compare(str2);

    //If the strings are equal, return zero.
    //If str1 is greater than str2, return value >0
    //If str2 is greater than str1, return value <0

    cout << Comp_result << endl;
  return 0;
}
```

String Functions

- <u>String searching</u>
- We can use the find() function to check whether a given character or a substring is present in the string or a part of string.

```
#include <iostream>
#include <string>
using namespace std;

int main () {
    string text = "C++ Programming";
    int position = text.find("Programming"); // Find the position of a substring

    cout << position << endl;
  return 0;
}
```

String Functions

- <u>Generating Substring</u>
- We can use the substr() function to generate a part of the string as a new string object.

```
#include <iostream>
#include <string>
using namespace std;

int main () {
    string text = "Hello, World!";
    string sub = text.substr(7, 5);
    //substr(position, length)

    cout << sub << endl;
  return 0;
}
```

String Functions

- Insert
- The insert() function not only allows us to add a string but also allows us to add it at the specified position.

```
#include <iostream>
#include <string>
using namespace std;

int main () {

    string text = "We love coding!";
    text.insert(7, " contest and");
    // Insert " contest and" at position 7

    cout << text << endl;
  return 0;
}
```

String Functions

- Replace
- The replace() function replaces the part of the string with the given other string.

```
#include <iostream>
#include <string>
using namespace std;

int main () {

    string text = "My hear is hurting.";
    cout << text << endl;
    text.replace(3, 4, "head"); // Replace(index, size, str2)
    cout << text << endl;
  return 0;
}
```

String Functions

- Erase
- The erase() function is used to remove a character or a part of the string.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
  string text = "Just a Normal Example!";
  cout << text << endl;


  text.erase(6, 7); // text.erase(start, size);
  cout << text;
  return 0;
}
```

String Functions

- Uppercase and Lowercase (conversion)

```
#include <bits/stdc++.h>

using namespace std;

int main() {
  // s1 is the string which is converted to uppercase
    string s1 = "abcde";

    // using transform() function and ::toupper in STL
    transform(s1.begin(), s1.end(), s1.begin(), ::toupper);
    cout<<s1<<endl;

    // s2 is the string which is converted to lowercase
    string s2 = "WXYZ";

    // using transform() function and ::tolower in STL
    transform(s2.begin(), s2.end(), s2.begin(), ::tolower);
   cout<<s2<<endl;
}
```

String Functions

- <u>String to int Conversion Using atoi()</u>

```
#include <bits/stdc++.h>
// C++ program to demonstrate the
// functioning of the atoi() function

using namespace std;


int main()
{
    char* str1 = "141dddd";
    char* str2 = "3.14";

    int res1 = atoi(str1);
    int res2 = atoi(str2);

    cout << "atoi(" << str1 << ") is " << res1 << "\n";
    cout << "atoi(" << str2 << ") is " << res2 << "\n";

    return 0;

}
```

String Functions

- String to int Conversion Using stoi()

```
#include <bits/stdc++.h>
// C++ program to demonstrate the
// functioning of the atoi() function

using namespace std;


int main()
{
  string str1 = "45";
  string str2 = "3.14159";
  char str3[] = "31337 geek";

  int myint1 = stoi(str1);
  int myint2 = stoi(str2);
  int myint3 = stoi(str3);

  cout << "stoi(\"" << str1 <<
          "\") is " << myint1 << '\n';
    cout << "stoi(\"" << str2 <<
            "\") is " << myint2 << '\n';
    cout << "stoi(\"" << str3 <<
            "\") is " << myint3 << '\n';
    return 0;
}
```

String Functions

- Remove Whitespace and Newline Characters

```
#include <bits/stdc++.h>
// C++ program to demonstrate the
// functioning of the atoi() function

using namespace std;


int main()
{
  string str = "Setting the example\nRaising up the bar\nShowing what is possible\nSo
others may go far\n";

    cout << "Before removing whitespace characters" << endl;
    cout << str << endl << endl;

    cout << "After removing whitespace characters" << endl << endl;
    for (int i=0; i<str.size(); i++)
    {
        if (!isspace(str[i]))
            cout << str[i];
    }

    return 0;
}
```

String Functions

- Find the last and first occurance of a character in the String

```
#include <bits/stdc++.h>

using namespace std;

int main() {
  string str = "Learn Competitive Programming at NDUB_coder!";
    size_t first = str.find('m'), last = str.find_last_of('m');
    //The size_t is chosen so that it can store the maximum size of a theoretically possible array or an
    object. On a 32-bit system size_t will take 32 bits, on a 64-bit one 64 bits.
    //The size_t type is frequently used in array indexing and loop counting because it makes it simple
    and secure to save the contents. The type size_t can never contain a negative value.

    //To put it simply, think of npos as no-position. As a return value, it is usually used to indicate
    that no matches were found in the string. Thus, if it returns true, matches were found at no
    positions (i.e., no matches)

    if (first != string::npos)
    {
         cout << "First occurrence of m is at index = " << first << "\n";
    }
    if (last != string::npos)
    {
        cout << "Last Occurrence of m is at index = " << last << "\n";
    }

    return 0;
}
```

String Functions

- Reverse the String

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
    string str = "Learn Competitive Programming at NDUB_coders!";
    string rev(str.rbegin(), str.rend());
    cout << "Reverse = " << rev << "\n";
    return 0;
}
```

String Functions

- Sort the String

```
#include <bits/stdc++.h>
using namespace std;

int main()
{
    string str = "Learn Competitive Programming at NDUB_coders!";
    sort(str.begin(), str.end());
    cout << "Sorted String = " << str << "\n";

    return 0;
}
```

Advanced String Algorithms

- String hashing using Polynomial rolling hash function
- Rabin-Karp Algorithm for Pattern Searching
- KMP Algorithm for Pattern Searching
- Z algorithm (Linear time pattern searching Algorithm)
- Suffix Array Pattern Searching
- Aho-Corasick Algorithm for Pattern Searching
- Finite Automata algorithm for Pattern Searching
- Boyer Moore Algorithm for Pattern Searching
- Manacher's Algorithm – Linear Time Longest Palindromic Substring

Solve this String Problems

- https://toph.co/problems/tags/string-Toph Problem Set
- https://onlinejudge.org/index.php?option=onlinejudge&Itemid=8&category=204- UVa Online Judge Problem Set
- https://codeforces.com/problemset?tags=strings-Codeforces Problem set
- Open an account in each website if you haven't already, submit and practice your code there.

- **Always remember, the more problems you solve by yourself, the better you will be in Competetive Programming.**
  - **There are no shortcuts for success.**

References:

1. https://www.javatpoint.com/cpp-math-functions
2. https://www.cs.mtsu.edu/~xyang/2170/libfunctions.html
3. https://www.w3schools.com/cpp/cpp_strings.asp
4. https://www.geeksforgeeks.org/strings-for-competitive-programming/